

Exercises for Bridging to S2



Key Features

- Bridging notes (QR code)
- A wide variety of question types:
 - True-or-false question
 - Multiple-choice question
 - Conventional question

Bridging Note



Bridging Exercise 1

Algebraic Expressions

(Refer to Book 1A Chapter 3)

True-or-false Questions

Determine whether each of the following is true or false. (1 – 5)

1. $y \times y \times y \times y = 4y$ True / False
2. $4 - (x - y)$ is identical to $4 - x + y$. True / False
3. If x is a number, then $2x$ must be an even number. True / False
4. If x is an integer, then $x + 2$ must be an odd number. True / False
5. If $T_n = 4n + 1$ is the general term of a sequence, then 7 is a term of the sequence. True / False

Multiple-choice Questions

6. $a \times a \times a \times (a + a) =$
 A. a^5 . B. $2a^4$. C. $a^4 + a$. D. $3a^4 + a$.
 Refer to HKCEE 2005 Paper 2 Q1
7. $(-y^4) - (-y)^4 =$
 A. 0. B. $2y^4$. C. $-2y^4$. D. $-y^8$.
8. $(-k)^3 - (-k^3) =$
 A. 0. B. $2k^3$. C. $-2k^3$. D. $-k^6$.
9. Let p be the smallest number of three consecutive odd number. The sum of the three numbers is
 A. $p + 2$. B. $p + 4$. C. $3p + 3$. D. $3p + 6$.
10. Let a_n be the n th term of a sequence. If $a_1 = 5$ and $a_{n+1} = a_n + n$ for any positive integer n , then $a_4 =$
 A. 8. B. 9. C. 11. D. 15.

Conventional Questions**Level 1**

Represent the following sentences or phrases in algebraic expressions. (11 – 18)

11. Subtract h from 52.

12. Add b to d .

13. The sum of $3a$ and $5b$

14. $9m$ minus $6n$.

15. Multiply $2x$ by $3y$.

16. a times $9b$.

17. Divide $5h$ by $11k$.

18. Divide $6p$ by $8q$.

19. Write the algebraic expression $p - 5 + 4q$ as a word sentence.

Simplify each of the following expressions. (20 – 25)

20. $r \div (8t) \times 4s$

21. $2m \times 15n \div 10$

22. $2 \times m + 8 \times m \div 4$

23. $3x - 7 - x + 2x$

24. $2x - 3y + 7y - 2x + y$

25. $6 - 2x - 5y + x - 7 - 4y$

Using the method of substitution, find the value of the remaining unknown in each of the following formulas. (26 – 29)

26. $y = 1 + 3x$ if $x = -2$

27. $y = 2 - 9x$ if $x = -3$

28. $h = -2k(3 - k)$ if $k = 1$

29. $h = -\frac{3k}{4k-1}$ if $k = -2$

In each of the following sequences, guess the next two terms and give an explanation. (30 – 31)

30. 12, 24, 36, 48, ...

31. 17, 21, 25, 29, ...

In each of the following, T_n is the general term of a sequence. Find the first three terms of the sequence. (32 – 33)

32. $T_n = 6 - n^2$

33. $T_n = \frac{24}{n+1}$

Level 2

Simplify each of the following expressions. (34 – 39)

34. $12x - 7y - 7xy - 2x + 7y$

35. $5t + 12s \div (2t) - 9s + 5t - 4s$

36. $-8x - y - (4x - 3y)$

37. $3(y - 5x) - 4(2x - 3y)$

38. $\frac{x}{3} + \frac{6y - x}{4}$

39. $\frac{6 + x - 2y}{8} - \frac{y - 3x + 1}{6}$

40. Four unknowns a , b , h and k are given. Represent the sentence ‘multiply the result of h minus k by a , and then divide the result by b ’ in an algebraic expression.

41. Consider the formula $R = \frac{u + v}{u - v}$.

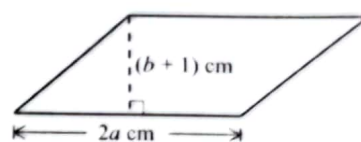
- (a) Find the value of R when $u = 9$ and $v = -3$.
- (b) Can the value of R be zero? Explain your answer.

42. Suppose that k is the largest number among three consecutive even numbers. Let S be the sum of the three consecutive even numbers.

- (a) Write down the formula for S .
(b) If the largest number among three consecutive even numbers is 36, find the sum of the three consecutive even numbers.

43. The figure shows a parallelogram. Let $A \text{ cm}^2$ be the area of the parallelogram.

- (a) Write down the formula for A .
(b) If $a = 4$ and $b = 3$, find the area of the parallelogram.



44. If a cake is cut into p equal parts, each part weighs 400 g.
(a) Express, in kg, the weight of the cake in terms of p .
(b) If the cake is cut into $4p$ equal parts, find the weight of each part in kg.



45. Coey has \$1 coins, \$2 coins and \$5 coins in her bag. The respective numbers of the three kinds of coins are p , q and r . Let T be the total amount of the coins in her bag.
- (a) Write down the formula for T .
 - (b) If $p = 7$, $q = 5$ and $r = 3$, find the total amount of the coins in her bag.

Challenging Questions

46. Let x and y be the units digit and the tens digit of a 2-digit number respectively. Let z be the value of the 2-digit number.
- (a) Write down the formula for z .
 - (b) Suppose that $x = 2$.
 - (i) Write down the formula for z .
 - (ii) Find all possible values of y such that the 2-digit number is divisible by 3.

47. It is given that the n th positive odd number N can be found by the formula $N = 2n - 1$.
- (a) Find the 6th positive odd number.
 - (b) If $N = 25$, find the value of n .
 - (c) Consider the sum S of the first n consecutive positive odd numbers $1, 3, 5, \dots, 2n - 1$. It is given that the sum S can be found by the formula $S = n^2$.
 - (i) Find the value of $1 + 3 + 5 + \dots + 25$.
 - (ii) Hence, find the value of $13 + 15 + 17 + \dots + 25$.

48. Consider the sequence: $1, 3, 4, 7, 11, \dots$.

It is given that for any positive integer n , the $(n + 2)$ th term is the sum of the $(n + 1)$ th term and the n th term.

$$\text{i.e., } T_3 = T_2 + T_1 = 3 + 1 = 4,$$

$$T_4 = T_3 + T_2 = 4 + 3 = 7,$$

$$T_5 = T_4 + T_3 = 7 + 4 = 11, \text{ and so on.}$$

Find the next three terms of the sequence.

Refer to HKCEE 2009 Paper 2 Q12

49. Consider the sequence: 3, 5, 9, 15,

It is given that for any positive integer n , $T_{n+1} = T_n + 2n$.

i.e., $T_2 = T_1 + 2(1) = 3 + 2(1) = 5$,

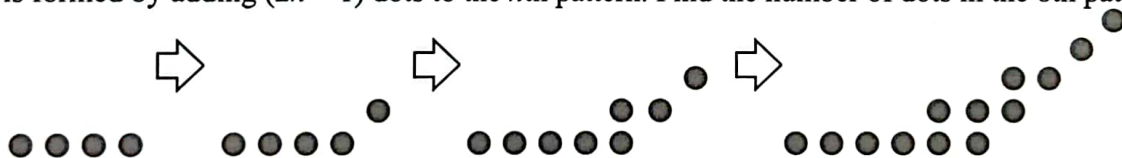
$T_3 = T_2 + 2(2) = 5 + 2(2) = 9$,

$T_4 = T_3 + 2(3) = 9 + 2(3) = 15$, and so on.

Find the next three terms of the sequence.

Refer to HKDSE 2012 Paper 2 Q12

50. In the figure, the 1st pattern consists of 4 dots. For any positive integer n , the $(n+1)$ th pattern is formed by adding $(2n-1)$ dots to the n th pattern. Find the number of dots in the 8th pattern.



Refer to HKDSE 2020 Paper 2 Q12

Bridging Note



Bridging Exercise 2

Linear Equations in One Unknown

(Refer to Book 1A Chapter 4)

True-or-false Questions

Determine whether each of the following is true or false. (1 – 3)

1. $3x - 8$ is an equation. True / False
2. The root of the equation $x = 21 - 6x$ is 3. True / False
3. The price of a pen is \$ x . The price of a book is \$5 more than 7 times the price of the pen. Their total price is \$69.
The equation $x + 7(x + 5) = 69$ can be used to find the value of x . True / False

Determine whether each of the following working steps is true or false. (4 – 5)

4. $4x - 3 = 11$
 $4x = 11 + 3$ True / False
5. $3(2 - x) - x = 2$
 $6 - x - x = 2$ True / False

Multiple-choice Questions

6. Which of the following equations has 4 as the solution?
A. $4 + x = 0$ B. $4 - x = 0$ C. $4x + 1 = 0$ D. $4x - 1 = 0$
7. Which of the following equations has 6 as the solution?
A. $x + 6 = 0$ B. $6x - 1 = 0$ C. $\frac{x}{6} - 1 = 0$ D. $\frac{x}{6} + 1 = 0$
8. Jason is x years old now. After 27 years, his age will be 4 times his present age. Which of the following equations can be used to find the value of x ?
A. $x + 27 = 4x$ B. $x - 27 = 4x$ C. $\frac{x}{4} + 27 = x$ D. $\frac{x}{4} - 27 = x$

9. Amy spends x hours on swimming every week. Tina spends 3 hours more than Amy on swimming every week. The total time they spend on swimming every week is 13 hours. Which of the following equations can be used to find the value of x ?
- A. $x + 3 = 13$ B. $x + 3x = 13$ C. $x + x + 3 = 13$ D. $x + x - 3 = 13$
10. A pack of *trading cards* is shared among 8 boys. Each boy can get x cards. If the pack of trading cards is shared among 9 girls, the number of trading cards each girl gets is less than that each boy gets by 2. Which of the following equations can be used to find the value of x ?
- A. $8x = 9(x - 2)$ B. $8x = 9x - 2$ C. $\frac{x}{8} = \frac{x}{9} + 2$ D. $\frac{x}{8} = \frac{x + 2}{9}$

Conventional Questions

Level 1

Solve the following equations. (11 – 28)

11. $9 - x = 12$

12. $-24 = -6x$

13. $\frac{x}{4} + 7 = 5$

14. $7 - 2x = -2$

15. $3(15 - x) = 21$

16. $\frac{x+8}{4} = -3$

17. $4(1 + 3x) = -32$

18. $-5(4 - 3x) = -50$

19. $2(x - 5) - 7 = -23$

20. $8 - 3(4 - 3x) = -13$

trading card 收藏卡

21. $4x = -8x$

22. $2x - 3 = 4x + 9$

23. $23x - 4(5 + 6x) = -13$

24. $7x - 2(6 - 2x) = 21$

25. $6(x - 3) = 5(2x - 6)$

26. $4(x + 3) + 2(3 - 4x) = 18$

27. $\frac{3z - 2}{4} = 3 - z$

28. $\frac{5z}{9} - \frac{2z}{7} = 17$

29. Christine bought 7 pears with a price of \$ x each. She paid \$50 and the change was \$15.

(a) Set up an equation in x .

(b) Find the price of a pear.

30. Nick wants to send a parcel and the postage fee is \$16. If he has five \$1.4 stamps, how many \$3 stamps should he buy?
31. This year, 7 times of Betty's age is equal to 3 times of her father's age. If she is 24 years younger than her father, find her father's age this year.

Level 2

Solve the following equations. (32 – 37)

32. $3(s - 0.7) = 5.7 - 0.6(1 - s)$

33. $\frac{2x+31}{5} - 4x = -1$

34. $\frac{x}{2} + \frac{x+3}{5} = 2$

35. $\frac{b+9}{7} + \frac{b-7}{3} = -2$

36. $8x - 2[3x - 3(1 - 2x)] = -24$

37. $4\left[3x - \frac{5(5-x)}{8}\right] = \frac{5(3x+7)-4}{2}$

38. There are only two kinds of tickets for a concert: regular tickets and premium tickets. The prices of a regular ticket and a premium ticket are \$200 and \$360 respectively. On a certain day, the number of regular tickets sold is 4 times the number of premium tickets sold and the sum of money for the tickets sold is \$139 200. Find the total number of tickets sold that day.

Refer to HKDSE 2017 Paper 1 Q4

Challenging Questions

39. (a) Solve the equation $4(x - 2) = 6x + 4$.

(b) Hence, if $4\left(\frac{a-1}{b} - 2\right) = \frac{6a-6}{b} + 4$, express a in terms of b .

40. In a *soccer league*, a team will get 3 points for a win, 1 point for a draw and no point for a loss. In last season, the Dragon Team played 22 matches. The number of draw matches was twice as many as the loss matches, and the number of win matches was 7 more than the draw matches.

(a) (i) How many loss matches did the team have last season?

(ii) How many points did the team get last season?

(b) In this season, 22 matches will be played. The team manager predicts that the number of win matches will not be more than last season. Can the team get more than 50 points this season? Explain your answer.



soccer league 足球聯賽

Bridging Note



Bridging Exercise 3 Polynomials

(Refer to Book 1A Chapter 5)

True-or-false Questions

Determine whether each of the following is true or false. (1 – 5)

1. $-\frac{m^4}{m^7}$ is equal to $-m^3$. True / False
2. The coefficient of the p^2 term of the polynomial $-p^4 - 2p^3 + p - 5$ is -2 . True / False
3. The degree of the polynomial $4x^3y^2 + 2xy - 3xy^5 + 6$ is 6. True / False
4. $-4x^5 - x^3 + 3x + 9$ is arranged in descending powers of x . True / False
5. When $k = -4$, the value of $-3k^3 + 2k^2 - 1$ is 159. True / False

Multiple-choice Questions

6. $\frac{(h^2k^4)^3}{h^8k^5} =$

A. $\frac{k^2}{h^2}$.

B. $\frac{k^2}{h^3}$.

C. $\frac{k^7}{h^2}$.

D. $\frac{k^7}{h^3}$.

Refer to HKDSE 2019 Paper 2 Q2

7. $\left(\frac{1}{8^{300}}\right)4^{200} =$

A. 1.

B. $\frac{1}{2^{100}}$.

C. $\frac{1}{2^{200}}$.

D. $\frac{1}{2^{500}}$.

Refer to HKDSE 2017 Paper 2 Q2

8. $(3s - 7s)s^5 =$

A. $-4s^5$.

B. $-4s^6$.

C. $-21s^6$.

D. $4s^7$.

Refer to HKDSE Practice Paper Paper 2 Q1

9. $(x^2 + 1)(x^2 - 2x - 1) =$

A. $x^4 - 1$.

B. $(x^2 - 1)^2$.

C. $x^4 - 2x^3 - 2x - 1$.

D. $x^4 - 2x^3 + 2x - 1$.

Refer to HKDSE 2015 Paper 2 Q1

10. Which of the following polynomials has two like terms?

A. $5x^2 - 5y^2$

B. $-x^3 - x^2 - x$

C. $8xy - 4yx$

D. $2x^3y - 3x^2 - 2xy - 1$

Conventional Questions**Level 1**

Represent each of the following expressions using index notation. (11 – 12)

11. $3 \times (-7) \times (-7) \times 3 \times (-7) \times 3$

12. $-k \times (-m) \times k \times (-m) \times k \times (-m) \times m$

Simplify each of the following expressions and write the answer in index notation. (13 – 24)

13. $(5^7)(5^2)$

14. $11(11)^{11}$

15. $e^5 \cdot e^6$

16. $b \cdot b^8$

17. $5q^3 \times 3q^5$

18. $u \times u^5 \times u^2$

19. $g^9 \div g^5$

20. $w^5 \div w^3$

21. $\frac{39x^8}{13x^3}$

22. $\frac{-4y^{11}}{20y^9}$

23. $k^{11} \div k^7 \div k$

24. $(8p^{20}) \div (-3p^2) \div (6p^5)$

36. $(3x + 2)(5x + 4)$

37. $(5 - 2x)(3x - 2)$

Simplify each of the following expressions. (38 – 39)

38. $x(-3 + 5y) + 4(5 - x)$

39. $5m(3m - 2n) - 4(2m^2 + 3mn)$

40. The area of square A is $x^2 \text{ cm}^2$. The length of a side of square B is $(x + 3) \text{ cm}$. If square B is larger, find the difference between the areas of the two squares.

Level 2

41. (a) Simplify $x + x + x + x$.
(b) Hence, simplify $(x + x + x + x)(x^2 + x^2 + x^2)$.

Refer to HKCEE 2007 Paper 2 Q3

Simplify each of the following expressions. (42 – 43)

42. $25a^6 \div (5a^3) \times 2a$

43. $\frac{36x^6y^7}{3x^2y \times 4y^4}$

Simplify each of the following expressions. (44 – 45)

44. $(4mn - 3m) + (m + 2n) - (3mn + 5m - 2n)$

45. $[5x - (2x - 4)](x - 7) - (2x - 5)(2x + 1)$

46. Arrange $m^3n - 7mn^4 + 6m^5n^2 - 2 - m^2n^3$ in

- (a) ascending powers of m ,
(b) descending powers of n .

47. Consider the polynomial $7x^3y^2 + 4x^5y - x^2y^5$.

- (a) Arrange the terms of the polynomial in ascending powers of y .
(b) Write down the coefficient of the term with the highest degree.
(c) Find the value of the polynomial when $x = -1$ and $y = 2$.

48. It is given that the lengths of sides of a triangle are $(4x + y)$ cm, $(3y - 2x)$ cm and $5y$ cm.

- (a) Express the perimeter of the triangle in terms of x and y .
(b) Suppose $y = 5$. If the perimeter of the triangle is 51 cm, find the value of x .

Challenging Questions

49. Simplify $(x - 3y)^2(2x + y) + 4xy(x - y)$ and arrange the terms in ascending powers of y .

50. It is given that n is an integer larger than 1.

- (a) Simplify $3^{2n+1} \times 3^{n-1}$.
(b) Hence, simplify $3^{2n+1} \times 3^{n-1} \div 3^{2n}$.

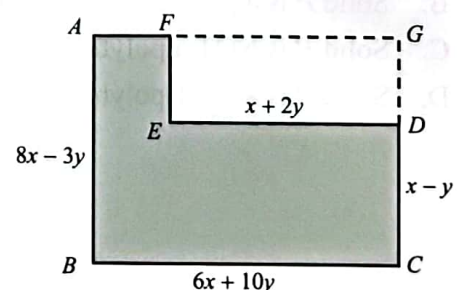
51. (a) Expand $x^2y^2(x^2 + y^2)$.
 (b) It is given that $xy = -6$ and $x^2 + y^2 = 13$. Find the value of $x^4y^2 + x^2y^4$.

52. (a) Expand $(x + y)(6x - y)$.
 (b) Using the result of (a), expand $(x + y + 3)(6x - y)$.

Refer to HKCEE 2011 Paper 2 Q3

53. Consider the polynomial $-3a^3b^4 + a^4b - 2a^2b^n$, where n is a positive integral constant.
- Arrange the terms of the polynomial in descending powers of a .
 - If the degree of the polynomial is 7, write down all possible values of n .
 - Suppose that the degree of the polynomial is 8.
 - Write down the value of n .
 - Arrange the terms of the polynomial in descending powers of b .
 - Write down the coefficient of the term with the lowest degree.
 - Find the value of the polynomial when $a = -2$ and $b = 1$.

54. In the figure, $ABCG$ and $DEFG$ are rectangles.
- Find the perimeter of the polygon $ABCDEF$.
 - Find the area of the polygon $ABCDEF$.



Bridging Note



Bridging Exercise 4 Area and Volume (1)

(Refer to Book 1A Chapter 6)

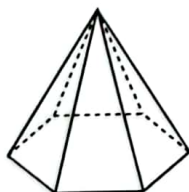
True-or-false Questions

Determine whether each of the following is true or false. (1 – 5)

- | | |
|---|---------------------|
| 1. A triangular pyramid has 4 faces. | <u>True / False</u> |
| 2. A cylinder has a uniform cross-section. | <u>True / False</u> |
| 3. A cross-section of a right triangular prism cannot be a rectangle. | <u>True / False</u> |
| 4. In a prism, a uniform cross-section must be identical to a base. | <u>True / False</u> |
| 5. The volume of a cube of side 2 cm is 16 cm^3 . | <u>True / False</u> |

Multiple-choice Questions

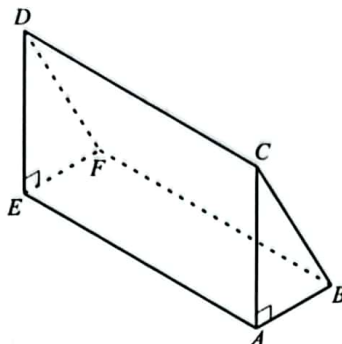
6. The figure shows two solids, P and Q .

Solid P Solid Q

Which of the following choices is correct?

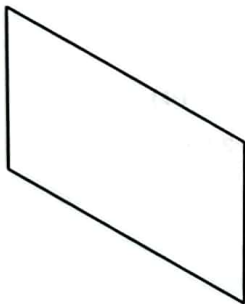
- A. Solid P is a polyhedron. Solid Q is a polyhedron.
 B. Solid P is a polyhedron. Solid Q is NOT a polyhedron.
 C. Solid P is NOT a polyhedron. Solid Q is a polyhedron.
 D. Solid P is a NOT polyhedron. Solid Q is NOT a polyhedron.

7. The figure shows a right prism. Its base is a triangle. Mike sketches a cross-section which is parallel to the plane $ACDE$.



Which of the following could express the plane diagram of the cross-section?

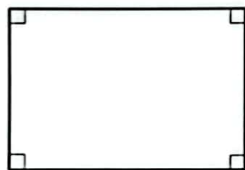
A.



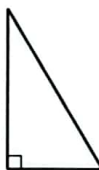
B.



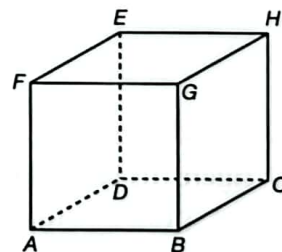
C.



D.



8. In the figure, $ABCDEFGH$ is a cube. Which of the following is the projection of the vertex A on the face $BCHG$?

A. B B. C C. D D. F 

9. The total surface area of a cube of side 3 cm is

A. 9 cm^2 .B. 36 cm^2 .C. 27 cm^2 .D. 54 cm^2 .

10. The length, the width and the height of a cuboid are 12 cm, 6 cm and k cm respectively. If the total surface area of the cuboid is 432 cm^2 , find the volume of the cuboid.

A. 432 cm^3 B. 360 cm^3 C. 576 cm^3 D. 1440 cm^3

Conventional Questions

Level 1

Each of the following figures is an incomplete diagram. Add solid line(s) and dotted line(s) in the figure so as to form a diagram of the specified solid. (11 – 12)

11. A cuboid



12. A pyramid with a triangular base



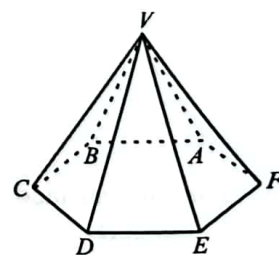
Use solid lines and dotted lines to sketch each of the following solids. (13 – 14)

13. A triangular prism

14. A pyramid with a square base

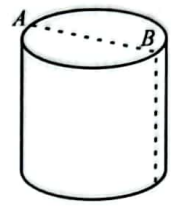
15. A right pyramid is placed horizontally as shown in the figure. Its base $ABCDEF$ is a regular hexagon.

- Sketch a cross-section which is parallel to the base.
- Sketch a cross-section which is perpendicular to the base and passing through the vertex V .

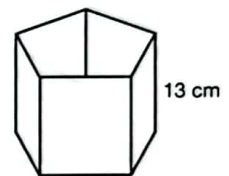


16. A right cylinder is placed horizontally as shown in the figure.

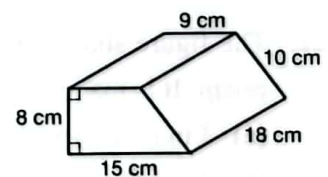
- (a) If it is cut vertically along the line AB , sketch the cross-section obtained.
- (b) Does the solid have any uniform cross-section? If it does, sketch the uniform cross-section(s).



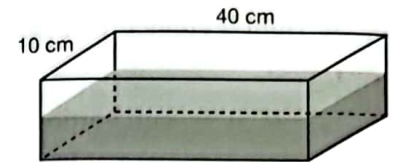
17. In the figure, the open box is in the shape of a right prism with a regular pentagonal base. It is made of paper. If the area and the perimeter of the base are 59.5 cm^2 and 30 cm respectively, find the total area of paper required to make the open box.



18. The figure shows a right prism. Its bases are trapeziums.
- (a) Find the volume of the prism.
- (b) Find the total surface area of the prism.



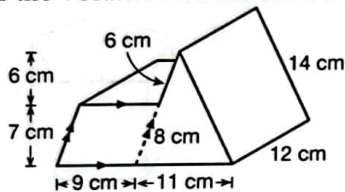
19. The figure shows a rectangular tank. Billy poured 8 glasses of water, each of volume 250 cm^3 , into the tank. Suppose that the water did not overflow. What is the rise in the water level?



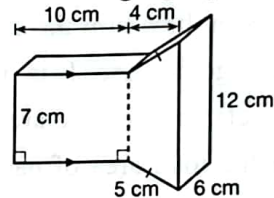
Level 2

Find the volume and the total surface area of each of the following right prisms. (20 – 21)

20.

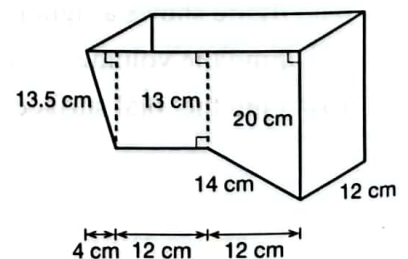


21.

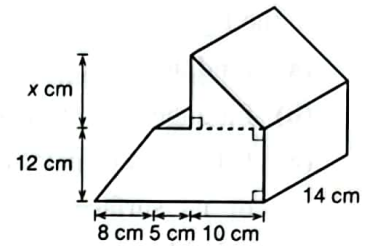


22. The figure shows an open container in the shape of a right prism. It is made of paper.

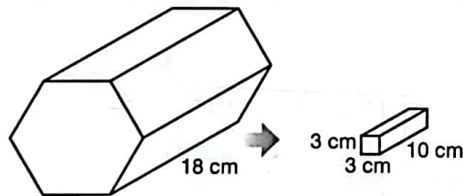
- Find the capacity of the container.
- Find the total area of the paper required to make the container.



23. In the figure, the volume of the right prism is 3962 cm^3 , find the value of x .

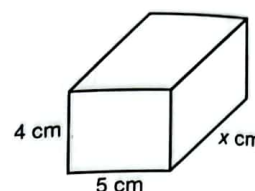


24. There is a solid metal block in the shape of a right hexagonal prism. Its base area is 420 cm^2 . A technician melted the metal block and recast it into some rectangular blocks.

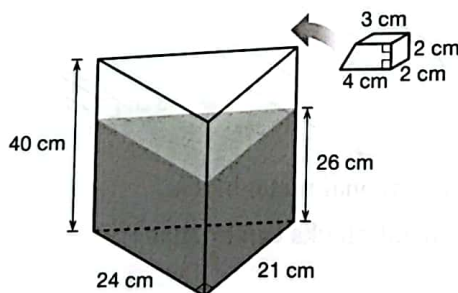


- (a) Find the volume of the hexagonal metal block.
 (b) How many rectangular metal blocks can be made?

25. In the figure, the volume of the cuboid is 140 cm^3 .
- Find the value of x .
 - Find the total surface area of the cuboid.
 - If the cost of painting the cuboid is \$0.05 for each cm^2 of its surface, find the cost of painting the whole cuboid.



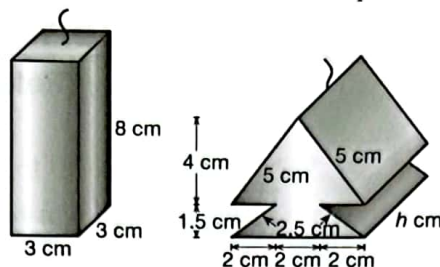
26. The figure shows a triangular tank. Ella puts 72 metal prisms with a trapezium base into the tank.



- Find the water level of the tank.
- After that, Ella puts more metal prisms into the tank. Find the maximum number of metal prisms can be put into the tank this time such that there is no overflow.

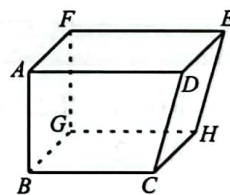
Challenging Questions

27. A rectangular candle is melted and recast into a tree-shaped candle as shown in the figure.



- (a) Find the value of h .
 (b) Find the change in the total surface area.

28. In the figure, $ABCDEFGH$ is a right prism. $ABCD$ is a trapezium and $AD \parallel BC$. It is given that $\angle BAD = 90^\circ$, $AD = 15$ cm, $BC = 12$ cm, $AF = 9$ cm and $CD = 5$ cm. Suppose that the volume of the prism is 486 cm³.



- (a) Find the length of AB .
 (b) Find the total surface area of the prism $ABCDEFGH$.

Refer to HKDSE 2012 Paper 1 Q9

Bridging Note



Bridging Exercise 5 Numerical Estimation

(Refer to Book 1B Chapter 7)

True-or-false Questions

Determine whether each of the following is true or false. (1 – 5)

1. In the decimal 6789.014, the place value of '7' is hundred. True / False
2. The integer 20 230 has 3 significant figures. True / False
3. The decimal 0.202 30 has 4 significant figures. True / False
4. The result of rounding off 389.56 to 3 significant figures is 390. True / False
5. The result of rounding off 0.091 345 to 2 decimal place is 0.091. True / False

Multiple-choice Questions

6. Round off 2.030 96 to 3 significant figures.
A. 2.03 B. 2.031 C. 2.0310 D. 2.03 000
7. Round off 0.089 96 to 3 decimal places.
A. 0.0899 B. 0.09 C. 0.090 D. 0.090 00
8. 0.086 520 1 =
A. 0.086 (*cor. to 2 sig. fig.*). B. 0.0865 (*cor. to 3 d. p.*).
C. 0.08652 (*cor. to 4 sig. fig.*). D. 0.086 520 (*cor. to 5 d. p.*).

Refer to HKDSE 2016 Paper 2 Q4

9. $\frac{3}{\pi^3} =$
A. 0.0967 (*cor. to 3 sig. fig.*). B. 0.096 75 (*cor. to 4 sig. fig.*).
C. 0.096 76 (*cor. to 5 d. p.*). D. 0.096 754 (*cor. to 6 d. p.*).

Refer to HKDSE 2017 Paper 2 Q4

10. The income of a shop yesterday was \$126 000, correct to 3 significant figures. Which of the following may be the actual income of the shop yesterday?
A. \$125 498 B. \$126 384 C. \$126 509 D. \$130 000

Conventional Questions

Level 1

11. (a) What is the place value of '1' in 8 905 413 620? _____

(b) Round off 8 905 413 620 to the nearest millions. _____

12. (a) What is the place value of '9' in 3.141 592 654? _____

(b) Round off 3.141 592 654 to 4 decimal places. _____

13. (a) Write down the number of significant figures of 0.002 047 0. _____

(b) Round off 0.002 047 0 to 3 decimal places. _____

(c) Round off 0.002 047 0 to 3 significant figures. _____

Round off each of the following numbers to the place indicated in brackets. (14 – 15)

14. 247.315 [unit] _____

15. 0.001 674 [3 decimal places] _____

Round off each of the following numbers to the numbers of significant figures indicated in brackets.

(16 – 21)

16. 5678 [1 and 2] _____

17. 13 428 [2 and 4] _____

18. 20 956 [2 and 3] _____

19. 0.0305 [1 and 2] _____

20. 0.005 207 9 [2 and 4] _____

21. 6.002 16 [1 and 3] _____

22. Express $\frac{31}{7}$ as a decimal,

(a) correct to 3 decimal places, _____

(b) correct to 4 significant figures. _____

23. Express $9\frac{6}{23}$ as a decimal,

(a) correct to 4 decimal places, _____

(b) correct to 3 significant figures. _____

24. (a) Round up 135.79 to 1 significant figure.
(b) Round off 135.79 to the nearest integer.
(c) Round down 135.79 to 1 decimal place.

Refer to HKDSE 2014 Paper 1 Q3

25. (a) Round up 743.8697 to the nearest hundreds.
(b) Round down 743.8697 to 2 decimal places.
(c) Round off 743.8697 to 2 significant figures.

Refer to HKDSE 2020 Paper 1 Q3

26. (a) Round off 6092.3907 to the nearest tens.
(b) Round up 6092.3907 to 3 significant figures.
(c) Round down 6092.3907 to 3 decimal places.

Refer to HKCEE 2011 Paper 1 Q4

27. The volume of a can of coke is labelled to be 315 mL. Find the volume of the coke, correct to the nearest

(a) 100 mL,

(b) 10 mL.

Level 2

28. Estimate the value of $636 \div 79$ by rounding up each number to the nearest tens.
29. Estimate the value of $481 \times 15.3 \div 59.7$ by rounding off each number to 2 significant figures.
30. The lengths of a road is 1.9 km. 70 rubbish bins are evenly distributed on the road, including two bins at the two ends of the road. Find the distance between two neighbouring rubbish bins in m, correct to
- (a) the nearest m, (b) 3 significant figures.

31. The weights of 5 packs of pork are 345.2 g, 280.7 g, 198.9 g, 246.8 g and 445.8 g. Estimate the average weight of a pack of pork by rounding off each weight to the nearest 10 g.
32. The length, width and height of a rectangular box are 88.5 cm, 24.7 cm and 58.3 cm respectively. Estimate the volume of the box in m^3 by
- (a) rounding off each measurement correct to the nearest 10 cm,
 - (b) rounding up each measurement correct to the nearest 10 cm,
 - (c) rounding down each measurement correct to the nearest 10 cm.

24. (a) Round up 135.79 to 1 significant figure.
(b) Round off 135.79 to the nearest integer.
(c) Round down 135.79 to 1 decimal place.

Refer to HKDSE 2014 Paper 1 Q3

25. (a) Round up 743.8697 to the nearest hundreds.
(b) Round down 743.8697 to 2 decimal places.
(c) Round off 743.8697 to 2 significant figures.

Refer to HKDSE 2020 Paper 1 Q3

26. (a) Round off 6092.3907 to the nearest tens.
(b) Round up 6092.3907 to 3 significant figures.
(c) Round down 6092.3907 to 3 decimal places.

Refer to HKCEE 2011 Paper 1 Q4

27. The volume of a can of coke is labelled to be 315 mL. Find the volume of the coke, correct to the nearest

(a) 100 mL,

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(a) the nearest m,

(b) 3 significant figures.

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32. The length, width and height of a rectangular box are 88.5 cm, 24.7 cm and 58.3 cm respectively. Estimate the volume of the box in m^3 by
- (a) rounding off each measurement correct to the nearest 10 cm,
 - (b) rounding up each measurement correct to the nearest 10 cm,
 - (c) rounding down each measurement correct to the nearest 10 cm.

33. Kelly wants to buy the following items in a shop.

Item	Unit price	Quantity needed
Orange juice	\$24.8 per bottle	2 bottles
Soft drink	\$17.3 per bottle	3 bottles
Water	\$9.8 per bottle	4 bottles

- (a) By rounding up the unit price of each item to the nearest dollar, estimate the total amount that Kelly should pay.
- (b) If Kelly has only three \$50 banknotes, does she have enough money to buy all the items? Use the result of (a) to explain your answer.

Refer to HKCEE 2008 Paper 1 Q7

34. A theatre has 23 rows of seat and each row has 52 seats. Using an appropriate strategy, judge whether the theatre has enough seats for 1000 people.

35. There are four kinds of lunch sets, *A*, *B*, *C* and *D*, sold in a tuck shop. The following table shows the unit price and the quantity sold of each lunch set today.

Lunch set	Unit price	Quantity sold
<i>A</i>	\$39	38
<i>B</i>	\$47	19
<i>C</i>	\$56	28
<i>D</i>	\$58	20

Using an appropriate estimation strategy, judge whether the total income from selling lunch set today exceed \$6000.

Challenging Questions

36. Water leaks from a water pipe. Rachel collects 120 *water droplets* dripped from the pipe. The volume of water is measured to be 48.2 cm^3 . Use appropriate estimation strategies to answer the following questions.
- (a) Estimate the volume of one water droplet.
 - (b) Suppose Rachel collects the 120 water droplets in 5 minutes.
 - (i) Estimate the volume of water leaked from the pipe in one year. (Give the answer correct to the nearest m^3 .)
 - (ii) Assume that the water fee is \$14 per m^3 . How much money is wasted in a year due to the leakage of the water pipe?

water droplet 水滴

Bridging Note



Bridging Exercise 6

Coordinate Geometry (1)

(Refer to Book 1B Chapter 8)

True-or-false Questions

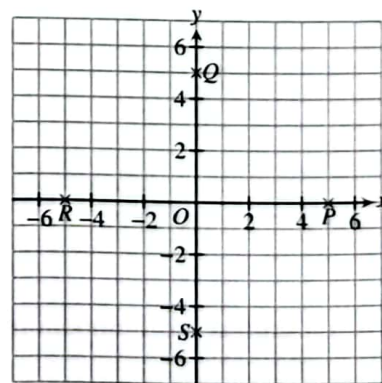
Determine whether each of the following is true or false. (1 – 5)

- The point $(-5, -3)$ lies in quadrant IV. True / False
- The line segment passing through the points $(3, 6)$ and $(3, -6)$ is vertical. True / False
- The length of the line segment joining the points $(-5, -2)$ and $(3, -2)$ is 8 units. True / False
- If the point $(4, -8)$ is reflected with respect to the y -axis, then the coordinates of its image are $(-4, -8)$. True / False
- If the point $(-2, 9)$ is rotate anti-clockwise about the origin through 90° , then the coordinates of its image are $(9, -2)$. True / False

Multiple-choice Questions

6. In the figure, which point can be represented by $(0, -5)$?

- A. P
 B. Q
 C. R
 D. S



7. If the point $(-7, 9)$ is rotated about the origin through 180° , then the coordinates of its image are

- A. $(-9, -7)$. B. $(9, 7)$. C. $(-7, -9)$. D. $(7, -9)$.

Refer to HKDSE Sample Paper Paper 2 Q26

8. If the point $(4, -7)$ is reflected with respect to the x -axis, then the coordinates of its image are

- A. $(-7, 4)$. B. $(-7, -4)$. C. $(-4, -7)$. D. $(4, 7)$.

9. L is a horizontal line passing through the point $(0, -6)$. If the point $(-3, -2)$ is reflected with respect to L , then the coordinates of the image are
- A. $(-9, -2)$. B. $(-3, -10)$. C. $(-3, 14)$. D. $(15, -2)$.

Refer to HKDSE Practice Paper Paper 2 Q25

10. The coordinates of the point A are $(-8, -5)$. A is translated rightwards by 12 units to the point B . B is then rotated clockwise about the origin through 90° to the point C . Find the y -coordinate of C .
- A. -5 B. -4 C. 4 D. 5

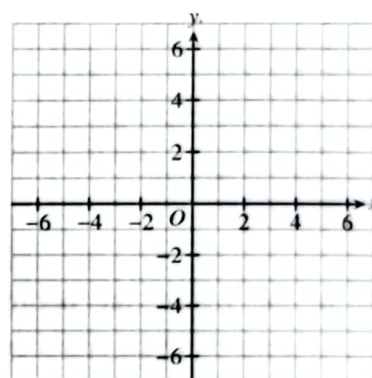
Refer to HKDSE 2019 Paper 2 Q25

Conventional Questions

Level 1

11. Four points $A(-4, -3)$, $B(4, -2)$, $C(5, 3)$ and $D(0, 4)$ are given.

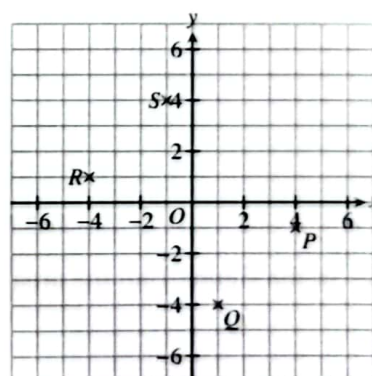
- (a) Plot the four points on the given rectangular coordinate plane. Then, join AC and BD .
- (b) Denote the point of intersection of AC and BD by P .
- (i) Write down the coordinates of P . _____
- (ii) In which quadrants does P lie? _____



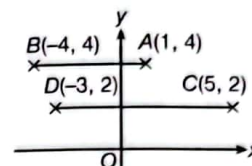
12. Refer to the figure.

- (a) Write down the coordinates of all the marked points.

- (b) Plot the points $A(1, 4)$, $B(4, 2)$ and $C(-3, 3)$ on the given rectangular coordinate plane.
- (c) A horizontal line passing through R intersects the y -axis at the point D . Plot the point D on the plane.
- (d) Consider the eight points A , B , C , D , P , Q , R and S . Determine which points lie in each of the following quadrants.
- (i) Quadrant I _____
- (ii) Quadrant II _____
- (iii) Quadrant III _____
- (iv) Quadrant IV _____

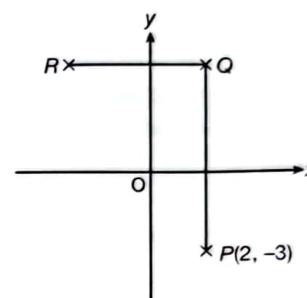


13. Find the lengths of line segments AB and CD in the figure.

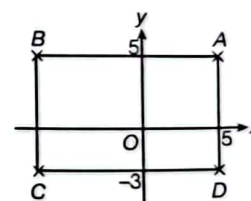


14. In the figure, PQ is a vertical line segment and QR is a horizontal line segment, their lengths are 7 units and 5 units respectively.

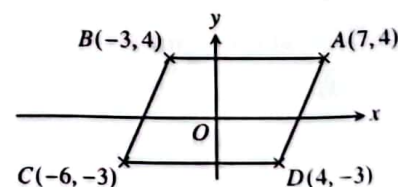
- (a) Find the coordinates of Q .
 (b) Find the coordinates of R .



15. In the figure, the perimeter of the rectangle $ABCD$ is 40 units. Find the coordinates of C .

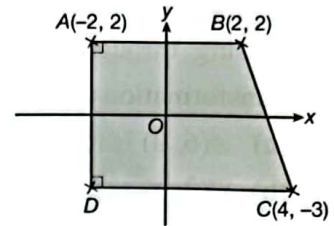


16. In the figure, find the area of the parallelogram $ABCD$.



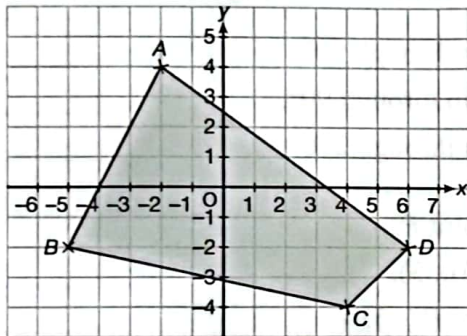
17. The figure shows a trapezium $ABCD$.

- (a) Find the coordinates of D .
 (b) Find the area of $ABCD$.

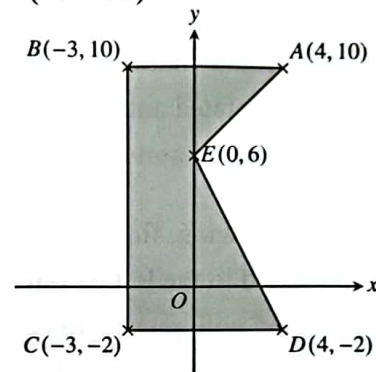


In each of the following, find the area of the shaded region. (18 – 19)

18.



19.



20. Find the coordinates of the images after the following transformations.

- (a) $A(-3, -5)$ is translated 8 units to the right.
 (b) $B(4, -9)$ is translated 3 units downwards.
 (c) $C(8, -6)$ is translated 11 units to the left.

21. Find the coordinates of the images after the following transformations.

- (a) $A(4, -5)$ is reflected with respect to the x -axis.
 (b) $B(-7, -3)$ is reflected with respect to the y -axis.
 (c) $C(5, 0)$ is reflected with respect to the x -axis.

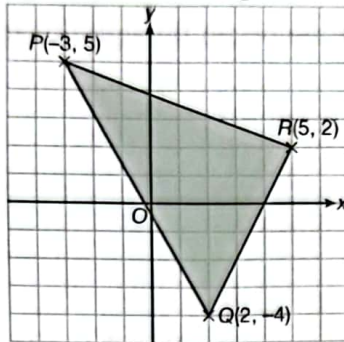
22. It is given that L_1 is a horizontal line passing through the point $(0, -2)$, L_2 is a vertical line passing through the point $(3, 0)$. Find the coordinates of the images after the following transformations.
- (a) $A(6, 0)$ is reflected with respect to the L_1 . _____
 - (b) $B(2, -4)$ is reflected with respect to the L_2 . _____
 - (c) $C(-6, -5)$ is reflected with respect to the L_1 . _____
23. Find the coordinates of the images after the following transformations.
- (a) $A(2, 3)$ is rotated clockwise about the origin through 90° . _____
 - (b) $B(4, -2)$ is rotated clockwise about the origin through 270° . _____
 - (c) $C(-7, 8)$ is rotated about the origin through 180° . _____
 - (d) $D(0, -3)$ is rotated anti-clockwise about the origin through 90° . _____
 - (e) $E(-3, -2)$ is rotated anti-clockwise about the origin through 90° . _____
 - (f) $F(6, -8)$ is rotated anti-clockwise about the origin through 270° . _____
24. In each of the following, find the coordinates of the point P .
- (a) P is translated to the left by 6 units to the point $P'(0, 3)$. _____
 - (b) P is reflected with respect to the x -axis to the point $P'(-5, 4)$. _____
 - (c) P is rotated anti-clockwise about the origin through 90° to the point $P'(-9, -5)$. _____

Level 2

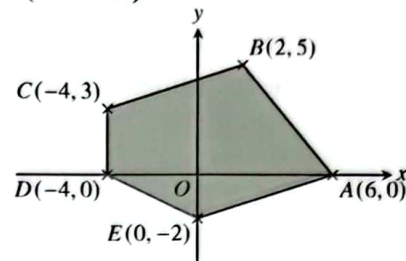
25. If the length of the line segment joining $A(4k, 2)$ and $B(6 - 2k, 2)$ is 6 units, find all the possible values of k .

In each of the following, find the area of the shaded region. (26 – 27)

26.



27.



28. It is given that $A(-2, 6)$, $B(-6, -2)$, $C(4, -2)$ and $D(4, 6)$ are the vertices of the polygon $ABCD$.

- What kind of polygon is $ABCD$?
- Find the area of the polygon $ABCD$.

29. It is given that $A(5, 0)$, $B(0, 4)$, $C(-3, -2)$, $D(-3, -5)$ and $E(0, -7)$ are the vertices of the polygon $ABCDE$. Find the area of the polygon $ABCDE$.

30. A point $A(3, 8)$ is translated 2 units to the right to the point B . B is then reflected with respect to the y -axis to the point C . Finally, C is rotated clockwise about the origin through 90° to the point D . Find the coordinates of D .
31. The coordinates of the points A and B are $(-4, 3)$ and $(5, 2)$ respectively. A' is the reflection image of A with respect to the y -axis. B is rotated anti-clockwise about origin O through 90° to the point B' .
- (a) Write down the coordinates of A' and B' .
 - (b) Describe how the point B' is transformed to A' only by translation.
32. The coordinates of the point A are $(-2, 3)$. A is rotated clockwise about the origin O through 90° to the point B .
- (a) Find the coordinates of B .
 - (b) Is $\triangle OAB$ an isosceles triangle? Explain your answer.

Refer to HKCEE 2011 Paper 1 Q8

33. Two points $A(-4, 2)$ and $B(0, -5)$ are rotated about the origin through 180° to the points C and D respectively. Find the area of the polygon $ABCD$.

Challenging Questions

34. The coordinates of the points A and B are $(2h - 7, 2)$ and $(h + 2, k - 3)$ respectively. A is translated to the right by 7 units and B is reflected with respect to the x -axis. It is given that the two images obtained are coincided.
- (a) Find the values of h and k .
 - (b) Find the coordinates of A and B .

Bridging Note



Bridging Exercise 7 Percentages (1)

(Refer to Book 1B Chapter 9)

True-or-false Questions

Determine whether each of the following is true or false. (1 – 5)

1. If x is more than y by 20%, then y is less than x by 20%. True / False
2. If x is double of y , then x is larger than y by 100%. True / False
3. If 100 is decreased to 20, then the percentage decrease is 20%. True / False
4. If the cost is \$100 and the selling price is \$200,
then the percentage profit is 100%. True / False
5. If the marked price is \$100 and the percentage discount is 30%,
then the cost must be \$70. True / False

Multiple-choice Questions

6. The cost of a bag is \$700. If it is sold at a loss of 40%, then the selling price is
A. \$280. B. \$420. C. \$500. D. \$980.
7. There are 378 members in a *recreation club*. If the number of male members is 25% less than that of female members, then the number of male members is
A. 162. B. 168. C. 210. D. 216.
Refer to HKDSE 2014 Paper 2 Q9
8. The cost of a car is $x\%$ higher than its selling price. After selling the car, the percentage loss is 20%. Find x .
A. 20 B. 25 C. 75 D. 80
Refer to HKDSE 2020 Paper 2 Q9
9. Jenny buys a skirt for \$720. She then sells the skirt to Kelly at a profit of 25%. At what price should Kelly sell the skirt in order to have a profit of 25%?
A. \$1080 B. \$1125 C. \$1200 D. \$1280
Refer to HKDSE Practice Paper Paper 2 Q10

recreation club 俱樂部

10. The monthly salary of Albert is 20% lower than that of Billy while the monthly salary of Billy is 20% higher than that of Calvin. It is given that the monthly salary of Albert is \$24 000. The monthly salary of Calvin is

A. \$16 000. B. \$23 040. C. \$25 000. D. \$37 500.

Refer to HKDSE 2016 Paper 2 Q10

Conventional Questions

Level 1

11. There were 140 customers in a canteen. 15% of them ordered set lunch *A*. Find the number of customers who did not order set lunch *A*.
12. In a hotel with 12 living floors, there are 30 rooms in each floor. If 65% of the rooms are rented, find the number of rooms that are not rented.
13. There are 63 oil paintings in an exhibition which is 42% of all the artworks. Find the total number of artworks.
14. Last month, a shop sold 76% of its computers with 18 computers left. Find the original number of computers in the shop last month.
15. The weight of a box of washing powder increases from 400 g to 450 g. Find the percentage increase in the weight.

16. The value of a stock decreased from \$75 to \$69. Find the percentage decrease.
17. The weight of Ada was 60 kg last month. Her weight decreased by 2% this month.
(a) Find the decrease in weight. (b) Find her weight this month.
18. Brian's income was \$12 000 last month. It is given that his income this month decreased by 4% when compared to last month. Find his income this month.
19. The airtime usage of Martin's mobile phone changed by -12% this month. Find his airtime usage last month if he uses 770 minutes this month.
20. Matthew sold a coin at \$468 and earned \$18. Find the percentage profit.
21. Betty sold a mobile phone at \$192 and lost \$608. Find the percentage loss.
22. Carmen sold a Hi-Fi set at \$976 and made a profit of 22%. Find the profit that she made.

23. Janice sold an LCD monitor at \$512 and suffered a loss of 20%. Find the loss.
24. Mr. Tong bought a book at \$130. He sold it to Mickey at \$156.
(a) Find the profit. (b) Find the percentage profit.
25. A ring is sold at \$3420 with a 24% loss. What is the cost of the ring?
26. A pair of earrings is sold at 15% off. If the marked price is \$560, find the discount of the pair of earrings.
27. An iron is sold at 25% off. If the selling price is \$498, find
(a) the marked price, and (b) the discount.

Level 2

28. There are 10 apples and 15 oranges in a bag. Then, 8 apples and x oranges are put into the bag such that the percentage of oranges in the bag becomes 64%.
(a) Find the value of x .
(b) Find the percentage change in the number of oranges in the bag. (Give the answer correct to the nearest 0.1%.)

29. A lorry is 60% longer than a minibus. By how many percents is the minibus shorter than the lorry?
30. The hourly wage of Peter increases from \$32 to \$40, while his daily working hour decreases by 20%. Find the percentage change in Peter's daily wage.
31. In a company, the number of female employees is 55% more than the number of male employees. The difference of the number of female employees and the number of male employees is 22. Find the number of female employees in the company.
Refer to HKDSE 2020 Paper 1 Q5
32. In a forum, there are 345 members and the number of male members is 30% more than the number of female members. Find the difference of the number of male members and the number of female members.
Refer to HKDSE 2016 Paper 1 Q5
33. The marked price of a toy is \$360. The toy is now sold at a discount of 20% on its marked price.
(a) Find the selling price of the toy.
(b) If the percentage profit is 44%, find the cost of the toy.
Refer to HKDSE 2014 Paper 1 Q6

34. A shirt is sold at a discount of 15% on its marked price. The selling price of the shirt is \$510.
- (a) Find the marked price of the shirt.
 - (b) After selling the shirt, the percentage profit is 50%. Find the cost of the shirt.

Refer to HKDSE 2019 Paper 1 Q5

35. In a shop, a bicycle is sold at 25% off such that a percentage profit of 20% is made. If the cost of the bicycle is \$425, find the marked price.

Refer to HKDSE Practice Paper Paper 1 Q4

36. The marked price of a necklace is 20% above its cost. A loss of \$56 is made by selling the necklace at a discount of 25% on its marked price. Find the marked price of the necklace.

Refer to HKDSE 2018 Paper 1 Q7

37. Miss Chan made a bottle of fruit punch. 25% of the volume was orange juice and 15% of the volume was lemon juice. The rest were soda water which contributed to 720 mL of the fruit punch.

- (a) Find the volume of the fruit punch.
- (b) If 300 mL of soda water was added, find the percentage of orange juice in the fruit punch.

- 38.** A shop sold 10 mobile phones at \$4400 each and a profit of 10% was made. It also sold 5 tablet computers at \$2720 each and a loss of 15% was made.
- (a)** Find the total cost of the 10 mobile phones and the 5 tablet computers.
 - (b)** The shop seller claims that a loss was made after selling the 10 mobile phones and the 5 tablet computers. Do you agree? Explain your answer.
- 39.** On a certain day, 20 000 visitors of a theme park were female. The number of male visitors was 30% less than that of female.
- (a)** Find the number of male visitors on that day.
 - (b)** On that day, 30 600 visitors of the park were tourists, while the rest were local citizens.
 - (i)** Find the percentage of tourists in the theme park on that day.
 - (ii)** It is given that 97% of the female visitors were tourists on that day.
 - (1)** Find the number of female tourists.
 - (2)** If $k\%$ of the male visitors were tourists, find the value of k .

Refer to HKCEE 2008 Paper 1 Q8

40. Suki has 60 stamps. The number of stamps owned by Alfred is 35% less than that owned by Suki.
- (a) How many stamps are owned by Alfred?
 - (b) Suppose that Suki gives x stamps to Alfred.
 - (i) Write down the number of stamps owned by Alfred in terms of x .
 - (ii) Write down the number of stamps owned by Suki in terms of x .
 - (iii) If Suki gives a certain number of stamps to Alfred, is it possible for them to have the same number of stamps? Explain your answer.

Refer to HKCEE 2010 Paper 1 Q7

41. In a jewellery shop, the cost of a necklace is 40% lower than its marked price. The necklace is sold at a 10% discount.
- (a) Let $\$c$ be the cost. Find the marked price and the selling price in terms of c .
 - (b) Can the shop make a profit in selling the necklace? Explain your answer.

Challenging Questions

42. A dog runs at a speed of 4 m/s. After completing 35% of the journey, it has to run 156 m to complete the journey.
- (a) Find the total length of the journey.
 - (b) How long does it take to complete the whole journey?
 - (c) If the journey is decreased by 15%, is the time taken also decreased by 15%? Explain your answer.
-
43. Suppose that the base and the height of parallelogram X are b cm and h cm respectively. Let A_1 cm² be the area of parallelogram X .
- (a) Write down the formula for A_1 .
 - (b) The base of parallelogram Y is 25% longer than that of parallelogram X , while the height of parallelogram Y is $p\%$ less than that of parallelogram X . Let A_2 cm² be the area of parallelogram Y .
 - (i) Write down the formula for A_2 in terms of b , h and p .
 - (ii) If the areas of parallelograms X and Y are the same, find the value of p .

Refer to HKCEE 2011 Paper 2 Q11

Bridging Note



Bridging Exercise 8

Identities

(Refer to Book 1B Chapter 10)

True-or-false Questions

Determine whether each of the following is true or false. (1 – 5)

1. $x^2 + 9 = 9 + x^2$ is an identity. True / False
2. $2(x + 3y) = 2x + 3y$ is an identity. True / False
3. $(2x + 5y)^2 = 4x^2 + 10xy + 25y^2$ is an identity. True / False
4. $(x + y)(-x - y) = -x^2 - 2xy - y^2$ is an identity. True / False
5. $(x - y)^2 + 2xy = x^2 + y^2$ is an identity. True / False

Multiple-choice Questions

6. Which of the following is an identity?
 A. $3(x - 6) = 3x - 6$ B. $\frac{x-6}{3} = x - 2$ C. $x - 6 = -6 + x$ D. $x - 6 = 0$
7. Which of the following is an identity?
 A. $x^2 - 9 = (x - 3)^2$ B. $x^2 - 9 = 0$
 C. $2(x^2 - 9) = 2x^2 - 9$ D. $(3 + x)(x - 3) = x^2 - 9$
8. $(x + 2y)\left(\frac{x}{2} - y\right) =$
 A. $x^2 - 2y^2$ B. $\frac{x^2}{2} - \frac{y^2}{4}$ C. $\frac{x^2}{2} - 2y^2$ D. $\frac{x^2}{4} - y^2$
9. $(u - v)(u + v - 5) =$
 A. $u^2 - v^2 - 5u - 5v$ B. $u^2 - v^2 + 5u + 5v$ C. $u^2 - v^2 + 5u - 5v$ D. $u^2 - v^2 - 5u + 5v$
10. If m and n are constants such that $(x - 6)(x + m) - 9 \equiv (x - 4)^2 + n$, then $n =$
 A. -109 B. -13 C. -9 D. -2

Refer to HKDSE 2019 Paper 2 Q4

Conventional Questions

Level 1

Determine whether each of the following equations is an identity. (11 – 14)

- | | | | |
|---------------------------------|-----------------|-----------------------------------|-----------------|
| 11. $x^2 + 2x + 1 = 0$ | <u>Yes / No</u> | 12. $x^2 + 25 = (x + 5)^2$ | <u>Yes / No</u> |
| 13. $36 - x^2 = (6 - x)(6 + x)$ | <u>Yes / No</u> | 14. $(8x - 1)^2 = 64x^2 - 8x + 1$ | <u>Yes / No</u> |

15. Prove that $4(-x + 2) + 2x = 3 - (2x - 5)$ is an identity.

16. Prove that $x(3x - 5) = 3(x^2 - x) + 2x$ is not an identity.

17. If P and Q are constants such that $Px + 8 \equiv 2(x - Q)$, find the values of P and Q .

18. If A , B and C are constants such that $(4x - 3)(2x + A) \equiv Bx^2 + Cx - 9$, find the values of A , B and C .

Expand the following expressions. (19 – 24)

19. $(2w - 3)(2w + 3)$

20. $(-4x - 3y)(-4x + 3y)$

21. $(-5c - 12w)(-5c + 12w)$

22. $-(-7 + 3x)^2$

23. $5(10x + 7y)^2$

24. $-2(7x - 4y)^2$

Evaluate the following without using a calculator. (25 – 28)

25. $98.8^2 - 101.2^2$

26. 30.3×29.7

27. -10.3^2

28. $(-19.8)^2$

Level 2

29. Prove that $(3x - 7)(7 + 3x) = 9(x^2 - 5) - 4$ is an identity.

Determine whether each of the following equations is an identity. (30 – 31)

30. $(x + 4)(3x - 2) = 3x(x + 2) + 4(x - 2)$

31. $(2x + 3)(5 - 2x) = 3x - (4x + 5)(x - 3)$

32. Let A , B and C be constants. If $6x^2 - Cx + 15 \equiv Ax(x - 2) + 5(x + B)$, find the values of A , B and C .

33. It is given that m and n are positive integers such that $m = 7 - 2n$.

(a) Complete the following table.

n	1	2	3
m			

(b) If k is a positive constant such that $2x^2 + 7x + k \equiv (2x + m)(x + n)$, write down all possible values of k .

Refer to HKCEE 2008 Paper 1 Q3

Expand the following expressions. (34 – 37)

34. $\left(\frac{1}{5}m - \frac{3}{5}n\right)\left(\frac{6}{7}n + \frac{2}{7}m\right)$

35. $4\left(-5z - \frac{3}{2}y\right)^2$

36. $6\left(\frac{x}{2} + \frac{3}{8}\right)^2$

37. $12\left(\frac{7x}{3} - \frac{5y}{9}\right)^2$

38. (a) Expand $(4x + y)^2$.

(b) Hence, expand $(4a + b - 2)^2$.

39. (a) Simplify $(x + y)^2 - 2xy$.
(b) It is given that $x + y = -8$ and $xy = 2$. Find the value of each of the following.
(i) $x^2 + y^2$ (ii) $(x - y)^2$

Challenging Questions

40. (a) Prove that $(x + y)^2 + (x - y)^2 \equiv 2(x^2 + y^2)$.
(b) Hence, find the value of each of the following.
(i) $199^2 + 201^2$ (ii) $98^2 + 99^2 + 100^2 + 101^2 + 102^2$

41. It is given that $x + \frac{1}{x} = 3$. Find the value of each of the following.

(a) $x^2 + \frac{1}{x^2}$

(b) $x^4 + \frac{1}{x^4}$

42. It is given that $x + y = 10$ and $xy = 9$, where $x > y$. Find the value of each of the following.
(a) $(x - y)^2$ (b) $x^2 - y^2$

Bridging Note



Bridging Exercise 9

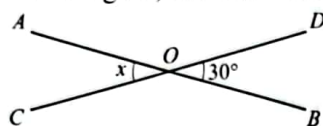
Angles and Parallel Lines

(Refer to Book 1B Chapter 11)

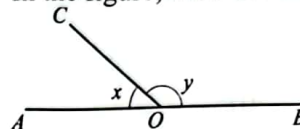
True-or-false Questions

Determine whether each of the following is true or false. (1 – 5)

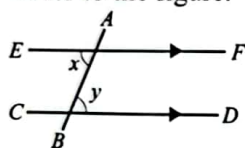
1. In the figure,
- COD
- is a straight line.

 x must be 30° .True / False

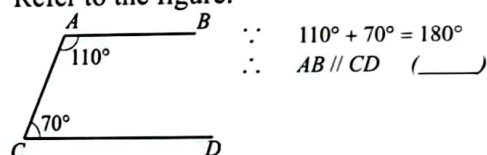
2. In the figure,
- AOB
- is a straight line.

 x and y must be complementary angles.True / False

3. Refer to the figure.

 x and y are corresponding angles.True / False

4. Refer to the figure.

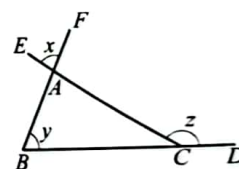


The reason in the above brackets is

int. $\angle$ s, $AB \parallel CD$.True / False

5. In the figure,
- BAF
- ,
- BCD
- and
- CAE
- are straight lines.

$z = x + y$

True / False

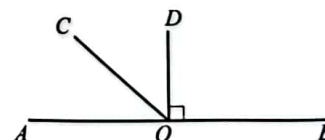
Multiple-choice Questions

6. The supplementary angle of
- 40°
- is

- A. 50° . B. 60° .
C. 140° . D. 260° .

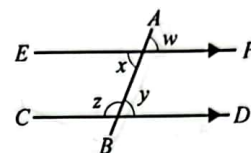
7. In the figure,
- AOB
- is a straight line. Which of the following is a right angle?

- A. $\angle AOB$ B. $\angle AOC$
C. $\angle AOD$ D. $\angle BOC$



8. In the figure, AB is a transversal of CD and EF . Which of the following is not true?

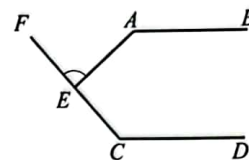
- A. $w = y$ B. $w = z$
C. $y + z = 180^\circ$ D. $x + z = 180^\circ$



9. In the figure, CEF is a straight line. If $AB \parallel CD$, $\angle BAE = 120^\circ$ and $\angle DCE = 130^\circ$, then $\angle AEF =$

- A. 50° . B. 60° .
C. 70° . D. 90° .

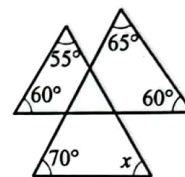
Refer to HKCEE 2005 Paper 2 Q28



10. In the figure, $x =$

- A. 50° . B. 55° .
C. 60° . D. 65° .

Refer to HKCEE 2001 Paper 2 Q20

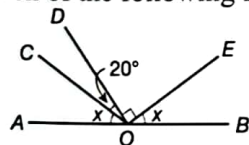


Conventional Questions

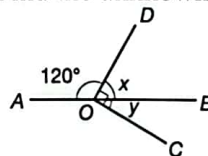
Level 1

In each of the following figures, AOB is a straight line. Find the unknowns. (11 – 12)

11.

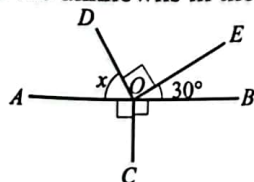


12.

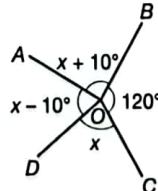


Find the unknowns in the following figures. (13 – 14)

13.

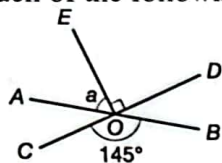


14.

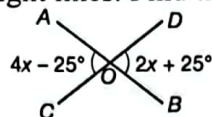


In each of the following figures, AOB and COD are straight lines. Find the unknowns. (15 – 16)

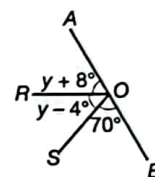
15.



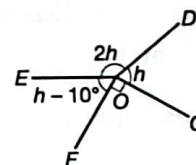
16.



17. In the figure, AOB is a straight line. Find y .

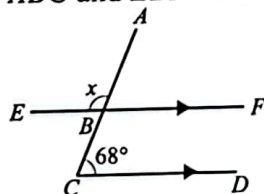


18. In the figure, find h .

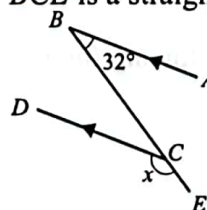


Find the unknowns in the following figures. (19 – 22)

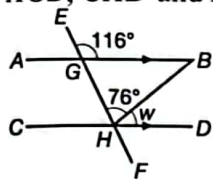
19. ABC and EBF are straight lines.



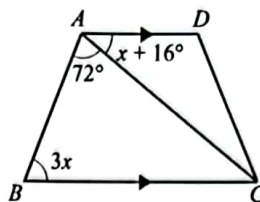
20. BCE is a straight line.



21. AGB , CHD and $EGHF$ are straight lines.

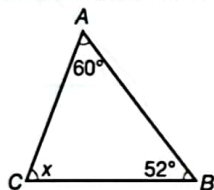


22.

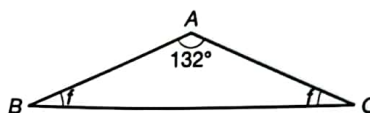


Find the unknowns in the following figures. (23 – 24)

23.

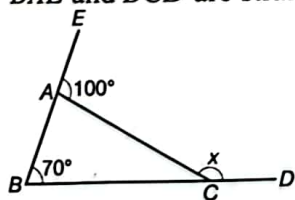


24.

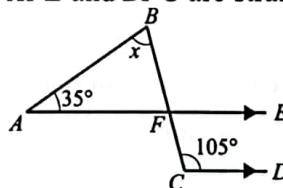


Find the unknowns in the following figures. (25 – 26)

25. BAE and BCD are straight lines.



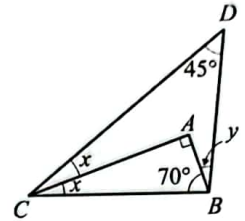
26. AFE and BFC are straight lines.



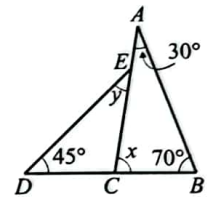
27. Refer to the figure.

(a) Find x .

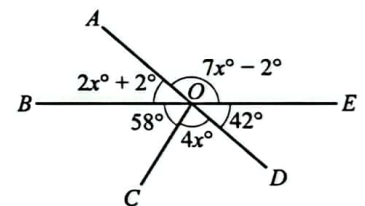
(b) Find y .



28. In the figure, AEC and BCD are straight lines. Find x and y .

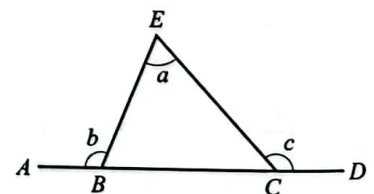


29. In the figure, is AOD a straight line? Explain your answer.



30. In the figure, $ABCD$ is a straight line.

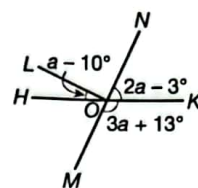
Prove that $b + c - a = 180^\circ$.



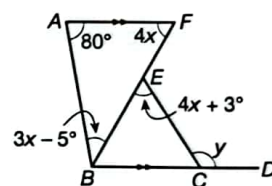
Level 2

31. In the figure, HOK and MON are straight lines.

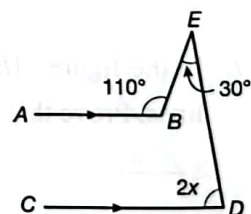
- (a) Find a .
 (b) Is OL perpendicular to MN ? Explain your answer.



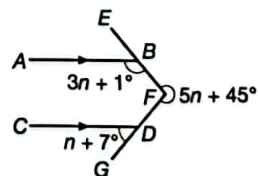
32. In the figure, BCD and BEF are straight lines. $AF \parallel BD$. Find x and y .



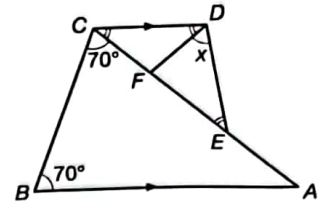
33. In the figure, $AB \parallel CD$. Find x .



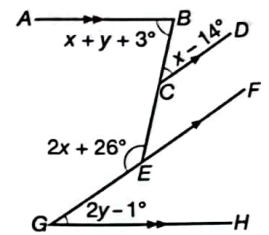
34. In the figure, $AB \parallel CD$. EBF and FDG are straight lines. Find n .



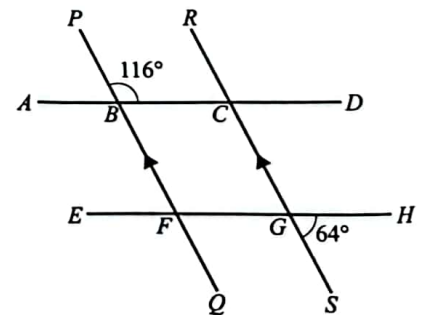
35. In the figure, $BA \parallel CD$. $AEFC$ is a straight line.
 $\angle DCF = \angle CDF = \angle DEF$. Find x .



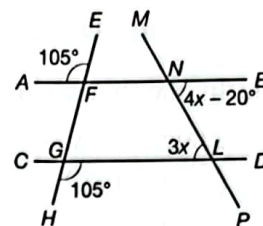
36. In the figure, BCE and FEG are straight lines. Find x and y .



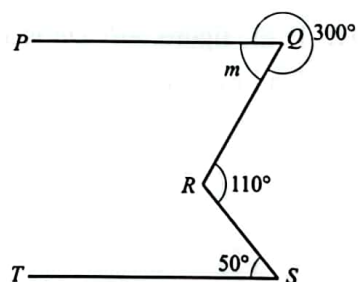
37. In the figure, $ABCD$, $EFGH$, $PBFQ$ and $RCGS$ are straight lines. Prove that $AD \parallel EH$.



38. In the figure, $AFNB$, $CGLD$, $EFGH$ and $MNLP$ are straight lines.
- (a) Prove that $AB \parallel CD$.
- (b) Find x .



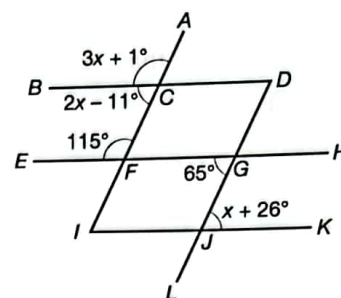
39. Refer to the figure.
- (a) Find m .
- (b) Prove that $PQ \parallel TS$.



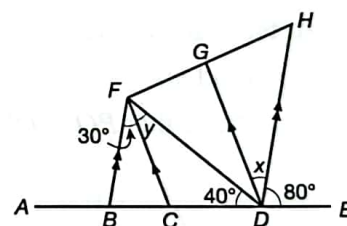
Challenging Questions

40. In the figure, BCD , $EFGH$, IJK , $ACFI$ and $DGJL$ are straight lines.

- (a) Find x .
 (b) Write down all pairs of parallel lines and give proofs to support your answers.

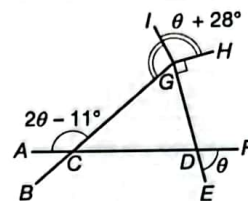


41. In the figure, $ABCDE$ is a straight line. $BF \parallel DH$, $CF \parallel DG$. Find x and y .

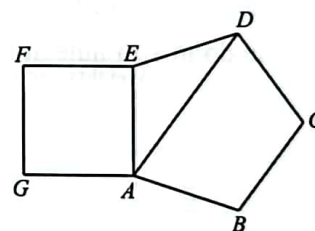


42. In the figure, $ACDF$, BCG and EDG are straight lines.
Find θ .

[Hint: θ is read as 'theta'. It is a Greek letter, which can be used to denote an unknown angle.]



43. In the figure, $AEFG$ is a square and $ABCDE$ is a regular pentagon. It is given that $\angle ADE = \angle EAD$ and $\angle ABC = 108^\circ$.
(a) Find $\angle EAD$ and $\angle BAG$.
(b) Prove that $AD \parallel BC$.
[Hint: The interior angles of a regular polygon are equal.]



Bridging Note



Bridging Exercise 10

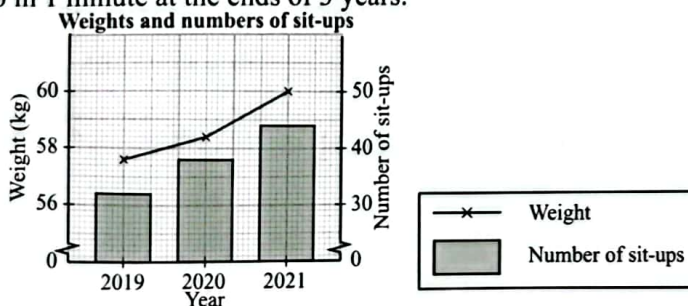
Statistical Charts (1)

(Refer to Book 1B Chapter 12)

True-or-false Questions

Determine whether each of the following is true or false. (1 – 5)

- The heights of students in a class are discrete data. True / False
- In the class interval 40 kg – 49 kg, the class mark is 45 kg. True / False
- In the class interval 71 cm – 75 cm, the upper class limit is 75 cm. True / False
- In the class interval 10 s – 19 s, the class width is 9 s. True / False
- The statistical chart shows the weights of Hugo and the numbers of sit-ups he can do in 1 minute at the ends of 3 years.



The weight of Hugo at the end of 2020 was 57.6 kg.

True / False

Multiple-choice Questions

- Which of the following are discrete data?
 - The numbers of books on a bookshelf
 - The thicknesses of books on a bookshelf
 - The heights of students in a room
 - The waiting time of a group of students
- In a set of data of integral values, the smallest and the largest data are 38 and 72 respectively. When a stem-and-leaf diagram is constructed, how many stem values are there?
 - 3
 - 5
 - 7
 - Cannot be determined

8. The stem-and-leaf diagram shows the distribution of the ages of teachers in a kindergarten.

Ages of teachers in a kindergarten										
Stem (10)	Leaf (1)									
2	2	k	5	5	5	5	6	7	7	8
3	0	1	2	3	4	5	6	6		
4	8									

How many possible values of k are there?

- A. 1 B. 2 C. 3 D. 4

(Questions 9 and 10)

The following frequency distribution table shows the waiting time (in min) of some passengers at a bus stop.

Waiting time (min)	1 – 5	6 – 10	?	?	?
Frequency	20	11	13	4	2

9. Find the class mark of the 4th class interval.
 A. 8 min B. 15.5 min C. 18 min D. 20.5 min
10. Find the percentage of passengers who wait not less than 15.5 min.
 A. 4% B. 8% C. 12% D. 38%

Conventional Questions

Level 1

11. The following data show the time (in minutes) between the schedules of two consecutive buses of 30 bus routes.

15	5	20	15	10	10	5	20	15	10
10	20	15	15	15	5	15	10	5	15
5	10	25	10	5	10	10	10	5	15

Complete the following frequency distribution table for the above data.

Time (minutes)	Tally	Frequency
5		
10		
15		
20		
25		

12. The following data show the weights (in kg) of school bags of 40 students.

6.2	6.1	6.3	5.3	5.7	6.9	5.3	6.4	6.2	5.8
5.9	7.1	7.1	6.7	6.6	5.8	7.2	5.7	6.3	6.6
6.4	5.4	5.0	5.9	6.5	6.1	6.5	5.4	5.3	5.8
5.4	6.3	6.0	5.6	6.4	7.0	6.3	6.6	7.2	7.1

Complete the following frequency distribution table for the above data.

Weight (kg)	Tally	Frequency
5.0 – 5.4		
5.5 – 5.9		

13. The following stem-and-leaf diagram shows the heights of a group of students.

Heights of a group of students	
Stem (10 cm)	Leaf (1 cm)
15	5 5 8 8
16	1 2 4 4 5 5 7
17	0 2 3 5 7 7 9 9 9
18	2 5 6
19	0 1

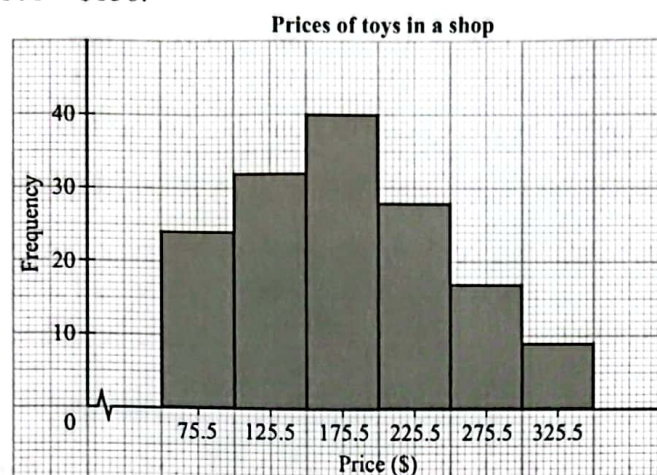
- (a) How many students are there in the group?
 (b) Find the percentage of students who are taller than 180 cm.

14. The frequency distribution table shows the scores of 40 students in a mathematics test.

Score	Number of students
21 – 40	3
41 – 60	11
61 – 80	21
81 – 100	5

- (a) Find the class mark of the 2nd class interval.
 (b) Find the class boundaries of the 3rd class interval.
 (c) How many students got not less than 60.5 marks?

15. The histogram shows the prices (in \$) of toys in a shop. The first two class intervals are \$51 – \$100 and \$101 – \$150.



- (a) Complete the following frequency distribution table.

Price (\$)	Class mark (\$)	Class boundaries (\$)	Frequency
51 – 100			
101 – 150			

- (b) Find the class width.
 (c) Find the least possible price of the toys in the class interval with the lowest frequency.
 (d) Find the percentage of toys with prices not less than \$200.5.

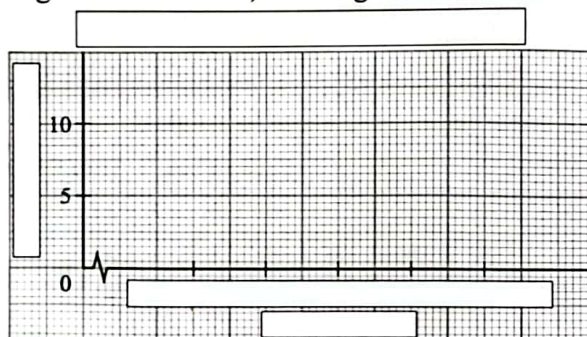
16. The following data show the heights (in cm) of 30 students in a class.

136	151	137	148	147	153	136	142	135	166
154	144	158	133	160	138	151	143	134	148
149	154	146	142	136	131	147	173	150	164

(a) Complete the following frequency distribution table for the data.

Height (cm)	Class mark (cm)	Frequency
151 – 160		

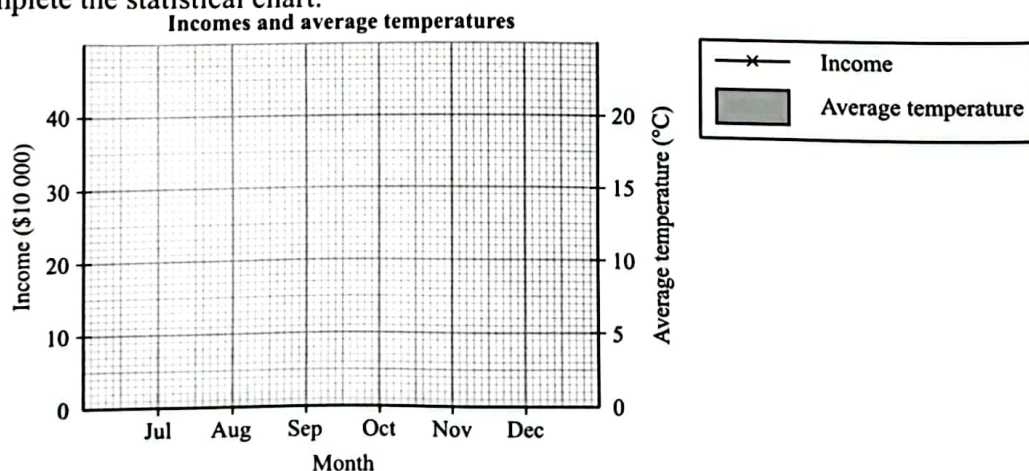
(b) Complete the histogram for the data, showing the class marks on the horizontal axis.



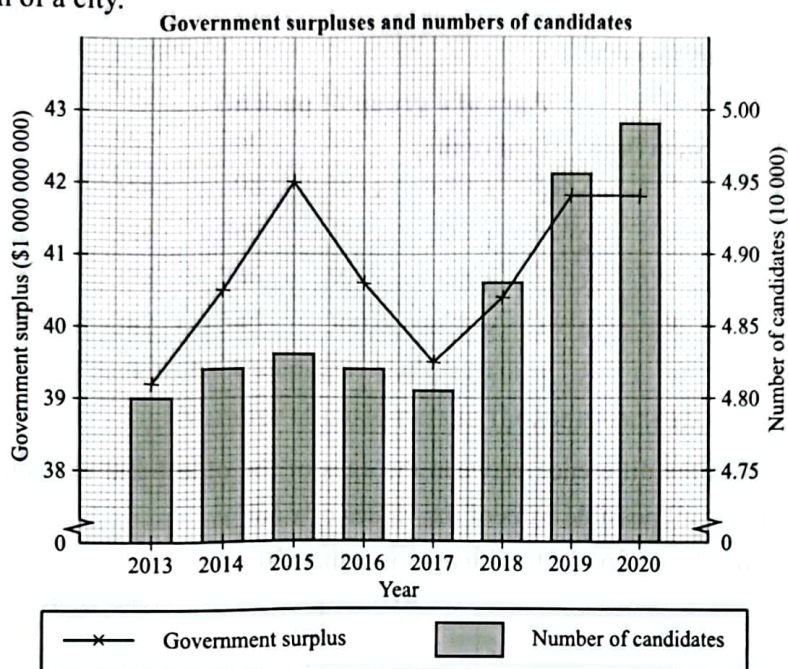
17. The following table shows the incomes of a winter sports company and the monthly average temperatures in the second half of last year.

Month	Jul	Aug	Sep	Oct	Nov	Dec
Income (\$10 000)	10	17	20	23	36	42
Temperature ($^{\circ}\text{C}$)	20.5	22	18	14.5	12.5	9

Complete the statistical chart.



18. The statistical chart shows the *government surpluses* and the numbers of candidates attending a public exam of a city.



- (a) In which year was the government surplus the least? Find the government surplus in this year.
- (b) In which year was the number of candidates the greatest? Find the number of candidates in this year.
- (c) Mike claims that the percentage increase in the government surplus from 2017 to 2020 is less than the percentage increase in the number of candidates from 2017 to 2020. Is the claim correct? Explain your answer.

government surplus 政府盈餘

Level 2

19. The stem-and-leaf diagram shows the ages of a group of patients, where x and y are non-negative integers less than 10.

Ages of a group of patients				
Stem (tens)	Leaf (units)			
1	x	4	5	8
2	0	1	4	
3	2	5		
4	3	8		
5	1	8	y	

- (a) Find the least possible value of y .
 (b) It is given that the difference between the ages of the oldest and the youngest patients is 48. Write down all possible pairs of the values of x and y .
 (c) Find the percentage of patients with ages less than 30.

Refer to HKDSE 2013 Paper 1 Q10

20. The following data show the heights (in cm) of 40 boys in S.1.

156	167	162	159	171	163	150	163	174	168
151	159	172	149	155	154	161	164	156	158
181	160	158	152	174	169	172	152	148	163
143	177	175	185	149	164	156	141	170	187

Complete the following frequency distribution tables.

Height (cm)	Tally	Frequency
135 – 154		
155 – 174		

Height (cm)	Tally	Frequency
140 – 149		
150 – 159		

21. The following data show the numbers of customers in two restaurants in the lunch period last month.

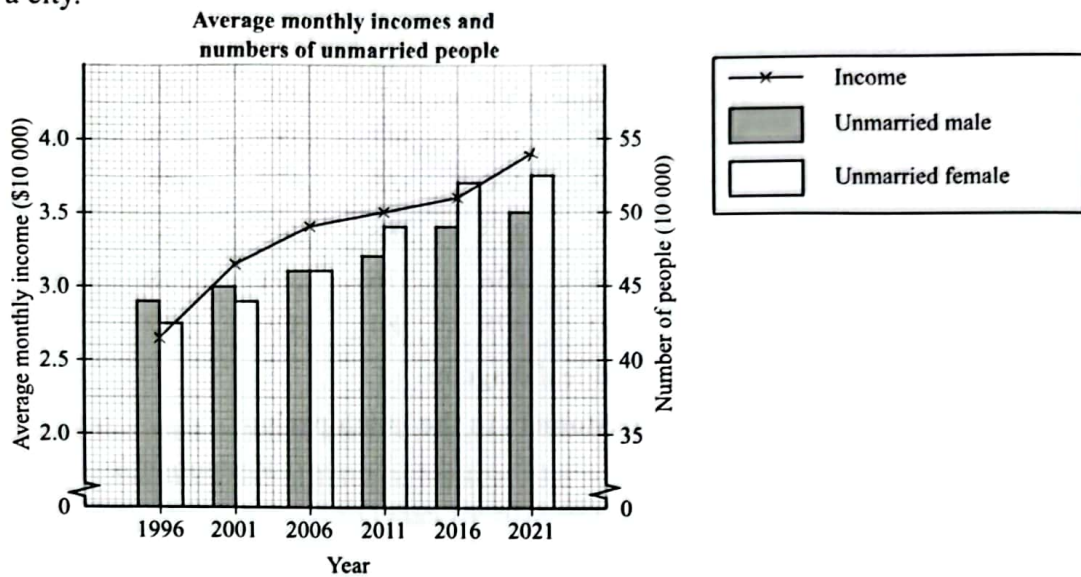
Restaurant <i>A</i>					Restaurant <i>B</i>				
340	347	342	355	342	352	396	395	387	363
350	357	362	360	392	385	347	383	356	382
353	387	354	398	383	390	348	377	378	393
370	364	346	349	375	353	388	345	363	333
379	343	378	346	304	358	376	391	378	393
368	352	382	345	360	378	384	339	378	324

- (a) Complete the back-to-back stem-and-leaf diagram.**

Number of customers in the lunch period		
<u>Restaurant A</u>		<u>Restaurant B</u>
Leaf (1)	Stem (10)	Leaf (1)
	30	
	31	

- (b) Which restaurant is more popular?**

22. The statistical chart shows the average monthly incomes and the numbers of unmarried people of a city.



- Which data, the number of unmarried males or the number of unmarried females, increased more rapidly?
- Find the percentage change in the average monthly income from 1996 to 2021. (Give the answer correct to 3 significant figures.)
- Find the percentage change in the total number of unmarried people from 1996 to 2021. (Give the answer correct to 3 significant figures.)

Answers

1. Algebraic Expressions (p. 2)

1. False 2. True 3. False
4. False 5. False 6. B
7. C 8. A 9. D
10. C 11. $52 - h$ 12. $d + b$
13. $3a + 5b$ 14. $9m - 6n$ 15. $6xy$
16. $9ab$ 17. $\frac{5h}{11k}$ 18. $\frac{3p}{4q}$
19. p minus 5 plus the product of 4 and q .
20. $\frac{rs}{2t}$ 21. $3mn$ 22. $4m$
23. $4x - 7$ 24. $5y$ 25. $-x - 9y - 1$
26. -5 27. 29 28. -4
29. $-\frac{2}{3}$ 30. 60, 72 31. 33, 37
32. 5, 2, -3 33. 12, 8, 6 34. $10x - 7xy$
35. $-13s + 10r + \frac{6s}{t}$ 36. $-12x + 2y$
37. $-23x + 15y$ 38. $\frac{x + 18y}{12}$
39. $\frac{15x - 10y + 14}{24}$ 40. $\frac{a(h - k)}{b}$
41. (a) $\frac{1}{2}$ (b) Yes
42. (a) $S = 3k - 6$ (b) 102
43. (a) $A = 2a(b + 1)$ (b) 32 cm^2
44. (a) $0.4p \text{ kg}$ (b) 0.1 kg
45. (a) $T = p + 2q + 5r$ (b) \$32
46. (a) $z = 10y + x$
- (b) (i) $z = 10y + 2$ (ii) 1, 4, 7
47. (a) 11 (b) 13
- (c) (i) 169 (ii) 133
48. 18, 29, 47 49. 23, 33, 45 50. 53

2. Linear Equations in One Unknown (p. 10)

1. False 2. True 3. False
4. True 5. False 6. B
7. C 8. A 9. C
10. A 11. -3 12. 4
13. -8 14. $\frac{9}{2}$ 15. 8
16. -20 17. -3 18. -2
19. -3 20. -1 21. 0
22. -6 23. -7 24. 3
25. 3 26. 0 27. 2
28. 63
29. (a) $50 - 7x = 15$ (b) \$5
30. 3 31. 42 32. 3

33. 2 34. 2 35. -2
36. 3 37. 4 38. 600
39. (a) -6 (b) $-6b + 1$
40. (a) (i) 3 (ii) 45
- (b) No

3. Polynomials (p. 16)

1. False 2. False 3. True
4. True 5. False 6. C
7. D 8. B 9. C
10. C 11. $-3^3 \times 7^3$ 12. k^3m^4
13. 5^9 14. 11^{12} 15. e^{11}
16. b^9 17. $15q^8$ 18. u^8
19. g^4 20. w^2 21. $3x^5$
22. $-\frac{y^2}{5}$ 23. k^3 24. $-\frac{4p^{13}}{9}$
25. (a) Yes (b) No (c) No
- (d) Yes (e) No (f) No
26. (a) -2 (b) 3 (c) $-2 - 5x + 3x^2 + 2x^3$
- (d) $2x^3 + 3x^2 - 5x - 2$
27. (a) 63 (b) 7 28. $-7w + 7$
29. $-11r + 7s$ 30. $9u^2 + 2u$ 31. $k^2 - 2$
32. $-28r + 36s$ 33. $35pq - 21q^2$
34. $x^2 - 2x - 35$ 35. $10x^2 + 49x - 5$
36. $15x^2 + 22x + 8$ 37. $-6x^2 + 19x - 10$
38. $5xy - 7x + 20$ 39. $7m^2 - 22mn$
40. $(6x + 9) \text{ cm}^2$
41. (a) $4x$ (b) $12x^3$ 42. $10a^4$
43. $3x^4y^2$ 44. $mn - 7m + 4n$
45. $-x^2 - 9x - 23$
46. (a) $-2 - 7mn^4 - m^2n^3 + m^3n + 6m^5n^2$
- (b) $-7mn^4 - m^2n^3 + 6m^5n^2 + m^3n - 2$
47. (a) $4x^5y + 7x^3y^2 - x^2y^5$ (b) -1
- (c) -68
48. (a) $(2x + 9y) \text{ cm}$ (b) 3
49. $2x^3 - 7x^2y + 8xy^2 + 9y^3$
50. (a) 3^{3n} (b) 3^n
51. (a) $x^4y^2 + x^2y^4$ (b) 468
52. (a) $6x^2 + 5xy - y^2$
- (b) $6x^2 + 5xy - y^2 + 18x - 3y$
53. (a) $a^4b - 3a^3b^4 - 2a^2b^n$
- (b) 1, 2, 3, 4, 5
- (c) (i) 6
- (ii) $-2a^2b^6 - 3a^3b^4 + a^4b$
- (iii) 1
- (iv) 32
54. (a) $28x + 14y$
- (b) $41x^2 + 50xy - 26y^2$

4. Area and Volume (1) (p. 22)

1. True 2. True 3. False
4. True 5. False 6. B
7. C 8. A 9. D
10. C 16. (b) Yes 17. 449.5 cm^2
18. (a) Volume = 1728 cm^3
(b) Total surface area = 948 cm^2
19. 5 cm
20. Volume = 1614 cm^3 , total surface area = 953 cm^2
21. Volume = 648 cm^3 , total area = 510 cm^2
22. (a) 4560 cm^3 (b) 1474 cm^2
23. 11
24. (a) 7560 cm^3 (b) 84
25. (a) 7 (b) 166 cm^2 (c) \$8.3
26. (a) 30 cm (b) 180
27. (a) 4 (b) $+22 \text{ cm}^2$
28. (a) 4 cm (b) 432 cm^2

5. Numerical Estimation (p. 30)

1. True 2. False 3. False
4. True 5. False 6. A
7. C 8. C 9. B
10. B
11. (a) Ten thousand (or 10 000)
(b) 8 905 000 000
12. (a) Hundred-thousandth (or 0.000 01)
(b) 3.1416
13. (a) 5 (b) 0.002 (c) 0.002 05
14. 247 15. 0.002
16. 6000, 5700 17. 13 000, 13 430
18. 21 000, 21 000 19. 0.03, 0.031
20. 0.0052, 0.005 208 21. 6, 6.00
22. (a) 4.429 (b) 4.429
23. (a) 9.2609 (b) 9.26
24. (a) 200 (b) 136 (c) 135.7
25. (a) 800 (b) 743.86 (c) 740
26. (a) 6090 (b) 6100 (c) 6092.390
27. (a) 300 mL (b) 320 mL
28. 8 29. 120
30. (a) 28 m (b) 27.5 m
31. 306 g
32. (a) 0.108 m^3 (b) 0.162 m^3
(c) 0.08 m^3
33. (a) \$144 (b) Yes
34. Yes 35. No
36. (a) 0.4 cm^3
(b) (i) 5 m^3 (ii) \$70

6. Coordinate Geometry (1) (p. 36)

1. False 2. True 3. True
4. True 5. False 6. D
7. D 8. D 9. B
10. B
11. (b) (i) (2, 1) (ii) I
12. (a) $P(4, -1)$, $Q(1, -4)$, $R(-4, 1)$, $S(-1, 4)$
(d) (i) A, B (ii) C, R, S
(iii) Nil (iv) P, Q
13. $AB = 5$ units, $CD = 8$ units
14. (a) (2, 4) (b) (-3, 4)
15. (-7, -3) 16. 70 sq. units
17. (a) (-2, -3) (b) 25 sq. units
18. 44 sq. units 19. 60 sq. units
20. (a) (5, -5)
(b) (4, -12)
(c) (-3, -6)
21. (a) (4, 5)
(b) (7, -3)
(c) (5, 0)
22. (a) (6, -4)
(b) (4, -4)
(c) (-6, 1)
23. (a) (3, -2)
(b) (2, 4)
(c) (7, -8)
(d) (3, 0)
(e) (2, -3)
(f) (-8, -6)
24. (a) (6, 3)
(b) (-5, -4)
(c) (-5, 9)
25. 0, 2
26. 28.5 sq. units
27. 44 sq. units
28. (a) Trapezium
(b) 64 sq. units
29. 48.5 sq. units
30. (8, 5)
31. (a) $A'(4, 3)$, $B'(-2, 5)$
(b) B' is first translated downwards by 2 units
and then to the right by 6 units.
32. (a) (3, 2)
(b) Yes
33. 40 sq. units
34. (a) $h = 2$, $k = 1$
(b) $A(-3, 2)$, $B(4, -2)$

7. Percentages (1) (p. 44)

1. False 2. True 3. False
4. True 5. False 6. B
7. A 8. B 9. B
10. C 11. 119 12. 126
13. 150 14. 75 15. 12.5%
16. 8%
17. (a) 1.2 kg (b) 58.8 kg
18. \$11 520 19. 875 minutes
20. 4% 21. 76%
22. \$176 23. \$128
24. (a) \$26 (b) 20%
25. \$4500 26. \$84
27. (a) \$664 (b) \$166
28. (a) 17 (b) +113.3%
29. 37.5% 30. 0%
31. 62 32. 45
33. (a) \$288 (b) \$200
34. (a) \$600 (b) \$340
35. \$680 36. \$672
37. (a) 1200 mL (b) 20%
38. (a) \$56 000 (b) No
39. (a) 14 000
- (b) (i) 90%
- (ii) (1) 19 400 (2) 80
40. (a) 39
- (b) (i) $39 + x$ (ii) $60 - x$ (iii) No
41. (a) Marked price = $\$ \frac{5c}{3}$, selling price = $\$ \frac{3c}{2}$
- (b) Yes
42. (a) 240 m (b) 60 s (c) Yes
43. (a) $A_1 = bh$
- (b) (i) $A_2 = 1.25(1 - p\%)bh$ (ii) 20

8. Identities (p. 53)

1. True 2. False 3. False
4. True 5. True 6. C
7. D 8. C 9. D
10. B 11. No 12. No
13. Yes 14. No 17. $P = 2, Q = -4$
18. $A = 3, B = 8, C = 6$
19. $4w^2 - 9$
20. $16x^2 - 9y^2$
21. $25c^2 - 144w^2$
22. $-9x^2 + 42x - 49$
23. $500x^2 + 700xy + 245y^2$
24. $-98x^2 + 112xy - 32y^2$
25. -480
26. 899.91

27. -106.09

28. 392.04

30. Yes

31. No

32. $A = 6, B = 3, C = 7$

(a)	n	1	2	3
	m	5	3	1

(b) 3, 5, 6

34. $\frac{2m^2}{35} - \frac{18n^2}{35}$

35. $100z^2 + 60zy + 9y^2$

36. $\frac{3x^2}{2} + \frac{9x}{4} + \frac{27}{32}$

37. $\frac{196x^2}{3} - \frac{280xy}{9} + \frac{100y^2}{27}$

38. (a) $16x^2 + 8xy + y^2$

(b) $16a^2 + 8ab + b^2 - 16a - 4b + 4$

39. (a) $x^2 + y^2$

(b) (i) 60 (ii) 56

40. (b) (i) 80 002 (ii) 50 010

41. (a) 7 (b) 47

42. (a) 64 (b) 80

9. Angles and Parallel Lines (p. 58)

1. False 2. False 3. False
4. False 5. True 6. C
7. C 8. B 9. C
10. A
11. 35° 12. $x = 60^\circ, y = 30^\circ$
13. 60° 14. 80° 15. 55°
16. 25° 17. 53° 18. 70°
19. 112° 20. 148° 21. 40°
22. 23° 23. 68° 24. 24°
25. 150° 26. 70°
27. (a) 20° (b) 25°
28. $x = 80^\circ, y = 35^\circ$
29. Yes
31. (a) 34° (b) No
32. $x = 15^\circ, y = 123^\circ$
33. 40°
34. 43°
35. 60°
36. $x = 56^\circ, y = 18^\circ$
38. (b) 20°
39. (a) 60°
40. (a) 38°
- (b) $BD \parallel EH, AI \parallel DL$
41. $x = 30^\circ, y = 30^\circ$
42. 75°
43. $\angle EAD = 36^\circ, \angle BAG = 162^\circ$

10. Statistical Charts (1) (p. 68)

1. False 2. False 3. True
 4. False 5. False 6. A
 7. B 8. D 9. C
 10. C

11.

Time (minutes)	Tally	Frequency
5	### //	7
10	### ###	10
15	### ///	9
20	///	3
25	/	1

12.

Weight (kg)	Tally	Frequency
5.0 – 5.4	### //	7
5.5 – 5.9	### ///	8
6.0 – 6.4	### ### //	12
6.5 – 6.9	### //	7
7.0 – 7.4	### /	6

13. (a) 25
 (b) 20%
 14. (a) 50.5
 (b) Lower class boundary = 60.5,
 upper class boundary = 80.5
 (c) 26

15. (a)

Price (\$)	Class mark (\$)	Class boundaries (\$)	Frequency
51–100	75.5	50.5 – 100.5	24
101–150	125.5	100.5 – 150.5	32
151–200	175.5	150.5 – 200.5	40
201–250	225.5	200.5 – 250.5	28
251–300	275.5	250.5 – 300.5	17
301–350	325.5	300.5 – 350.5	9

- (b) \$50
 (c) \$300.5
 (d) 36%

16. (a)

Height (cm)	Class mark (cm)	Frequency
131 – 140	135.5	9
141 – 150	145.5	11
151 – 160	155.5	7
161 – 170	165.5	2
171 – 180	175.5	1

18. (a) 2013; \$39 200 000 000
 (b) 2020; 49 900
 (c) No

19. (a) 8
 (b) $x = 0, y = 8; x = 1, y = 9$
 (c) 50%

20.

Height (cm)	Tally	Frequency
135 – 154	### ###	10
155 – 174	### ### ### ### ###	25
175 – 194	###	5

Height (cm)	Tally	Frequency
140 – 149	###	5
150 – 159	### ### ///	13
160 – 169	### ### /	11
170 – 179	### ///	8
180 – 189	///	3

21. (b) Restaurant B
 22. (a) Number of unmarried females
 (b) 47.2%
 (c) 18.5%